



DATA SCIENCE PROJECT LIFECYCLE

5DATA004C

INDIVIDUAL COURSEWORK

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Dataset: Internal displacements (new displacements) associated with disasters

Video: https://drive.google.com/file/d/1VILwTeiUCL_UzqUplCj3fRxRU6k3bxNz/view?usp=sharing

Streamlit: <https://disasterdisplacementanalysisdashboard.streamlit.app/>

GitHub: https://github.com/ShamlaAqthar19/w2149479_Disaster_Displacement_Dashboard

Aims and Objectives

Dataset: Internal displacements (new displacements) associated with disasters

Dataset Source: Humanitarian Data Exchange (HDX)

Source link: <https://data.humdata.org/dataset/idmc-internal-displacements-new-displacements-associated-with-disasters>

The dataset contains global records of disaster-related internal displacement events, hazard types, affected countries and displacement figures. The aim of this project is to develop an interactive data dashboard that analyses global disaster displacement trends, enabling users to better understand the scale, patterns and key drivers of human displacement based on climate and environmental risks

Objectives

- To identify patterns which include the most contributing hazard type and calculating displacements across time through exploratory data analysis.
- To clean and preprocess a large-scale dataset using python, addressing missing values, inconsistencies and formatting issues to ensure data reliability.
- To develop a dynamic and user-friendly visualisation using Plotly allowing the users to explore yearly trends
- To derive meaningful insights on displacement patterns hazard impacts and affected regions to support data driven understanding.
- To deploy the dashboard using Streamlit which ensures accessibility and enables real-time interaction for users.

The key insights selected for this dashboard were chosen to provide a comprehensive view of global humanitarian risks. Total displacement provides a clear overview of the overall severity of the issue while identifying the most contributing hazard helps understand the environmental drivers and climate related risks behind displacement. Timeseries trends show how displacement patterns have evolved over time, highlighting potential increases linked to climate change. Country-based analysis indicated geographical vulnerability and supports targeted decision making and resource allocation in high-risk regions.

These insights are aligned with UN SDG 13(climate action) and SDG 11(sustainable cities and communities) which focus on addressing climate risks and building resilient communities. Overall, the dashboard transforms raw data into meaningful insights for data driven decision making such as disaster preparedness and long-term sustainability planning.

Development Methodology

An iterative development approach was used where the dashboard was developed in incremental stages rather than in a single continuous process. Alternative methods such as waterfall were considered, however they were not suitable due to the need of continuous testing and adjustments during data exploration. Iterative methodology was selected because it allowed early validation of data processing and visual outputs, reducing the risk of errors in later stages.

It started off with loading the data, identifying key indicators and countries affected and then the main trend visualisations which involved using key variables such as year, country_name, hazard_type_name and new_displacement from the dataset. This initial stage was to verify data correctness and ensure that key metrics such as total displacement was accurately calculated.

The dashboard was then adjusted through multiple iterations where additional features such as hazard-based filtering, multiline trend analysis and geospatial visualisations were introduced. Each iteration included testing and debugging to ensure that new components did not interfere with the existing functionality.

This approach also allowed flexibility in handling real world issues such as missing values, data inconsistencies and layout improvements. Functional testing was conducted before final user interface enhancements, ensuring that the system remained accurate and stable before improving usability and presentation.

The development followed the key phases of the iterative model such as requirement analysis, system design, implementation, testing and evaluation which were revisited across multiple cycles.

Phase-1: Environmental Setup

Set up the development environment using Miniconda and installed required libraries such as Pandas, Plotly and Streamlit.

Phase-2: Initial data profiling

Initial exploration of the raw dataset was carried out to understand the data structure, identify missing values and data inconsistencies. This step provided an overview of data quality before cleaning.

Phase-3: Data Preparation

The dataset was cleaned and structured by handling missing values, removing unnecessary columns, duplicate rows and converting data formats where necessary.

Phase-4: Exploratory Data Analysis (EDA)

Performed data explorations to understand the identify key patterns such as most contributing hazard type and displacements overtime.

Phase-5: Dashboard Development

Developed an interactive dashboard using Streamlit with KPIs, line charts, bar charts and map visualisations. Filters were also implemented to allow users to explore the data dynamically.

Phase-6: Testing and Deployment

Functional testing was conducted using defined test cases to ensure accuracy and responsiveness. The finalised dashboard was then deployed on Streamlit for user access.

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Requirements

Functional	Non-Functional
The system shall automatically clean and structure the raw dataset upon execution handling missing values and inconsistencies to ensure data integrity before visualisations	The dashboard must have a good layout with clear visual hierarchy
The system shall display aggregated KPI metrics of total displacement, leading hazard types and the most affected country	Data loading must be optimized using <code>@st.cache_data</code> to ensure smooth interactivity.
The system shall provide integrated charts such as time series for displacements and top 5 hazards and bar charts for affected countries and displacements by hazard types to allow simultaneous trends and categorical analysis.	The system must provide a fast interaction with less latency.
The system shall include a map to illustrate the regional impact of disasters	The visualisation must be clear, readable and clutter free.
The app will provide filters and slicers to allow user interaction	

Test cases

#01	TC1	Title: : Global year filter functionality	Test case Title: ear range filtering validation
Description		Changing the year slider to a specific range to ensure all visuals adjust accordingly	
Steps and input data		Locate the ‘Years’ slider on the left-hand side of the dashboard Drag the lower bound to 2014 and upper bound to 2020 Observe the KPI and visuals	
Dependencies		cleaned_data.csv must be loaded Python environment must have pandas, plotly and streamlit libraries installed Streamlit sidebar must be functional	
Expected result		The total displaced KPI must reflect the sum for 2014-2020 The x axis of the line chart must display years from 2014 – 2020 No data point should be visible from 2008-2013 and 2021-2024	

#02	TC2	Title: Hazard type filter functionality	Test case Title: Single hazard selection validation
Description		Deselecting all hazard type and selecting on only ‘flood’ to analyse that specific hazard type	
Steps and input data		Locate the ‘Hazard Type’ multi selector on the left-hand side of the dashboard Deselect all the hazard types Choose only ‘flood’ as hazard type Observe the KPI and visuals	
Dependencies		cleaned_data.csv must be loaded Python environment must have pandas, plotly and streamlit libraries installed Streamlit sidebar must be functional	
Expected result		The total displaced KPI and most affected country must reflect the sum for floods	

	No other hazard types but flood must be visible in the legend of trend of top 5 hazard types and displacements by hazard type bar chart.
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#03	TC3	Title: top 5 Hazard trend logic	Test case Title: Top 5 Hazard display validation
Description		Ensuring that the multiline chart displays the top 5 hazards only to maintain the clarity of the dashboard.	
Steps and input data		Ensure that all hazard types are selected in the filter sidebar Observe the Hazard type trend chart Count the number of active lines in the legend	
Dependencies		The data aggregation logic must correctly use ‘.nlargest(5)’ on the filtered data frame. Python environment must have pandas, plotly and streamlit libraries installed Streamlit sidebar must be functional	
Expected result		Only 5 hazard types are displayed in the legend The chart is readable without lines overlapping The lines are continuous across the year range	

#04	TC4	Title: Geospatial map interaction	Test case Title: Map hover information validation
Description		Hover over a specific country on the map	
Steps and input data		Navigate to the global map of the dashboard Hover over any country on the map Observe tooltip information	
Dependencies		The choropleth function must correctly use ‘.hover_name’ on the filtered data frame cleaned_data.csv must be loaded Python environment must have pandas, plotly and streamlit libraries installed	
Expected result		The country name and total displacement must be displayed when hovering over the specific country	

#05	TC5	Title: KPI accuracy validation	Test case Title: KPI and chart consistency check
Description		Check the most affected country against the top 10 affected countries in the bar chart	
Steps and input data		Select a specific year range Select all hazard types Compare the KPI value and most affected country with the top 10 bar-chart	
Dependencies		cleaned_data.csv must be loaded python environment must have pandas, plotly and streamlit libraries installed Streamlit sidebar must be functional	
Expected result		The KPI value must match the first value in the Top 10 bar-chart	

Test log

TC	Date	Executed by	Actual result	Pass/Fail	Notes
01	21/04/2026	Shamla Aqthar	Dashboard successfully filtered. The KPI updated instantly. Charts correctly started at 2014.	Pass	
02	21/04/2026	Shamla Aqthar	Selecting 'flood' removed all other lines and bars. Top hazard type displayed correctly as 'Flood'	Pass	
03	22/04/2026	Shamla Aqthar	The system raised a 'ValueError'. The chart failed to display due to an error in how data was grouped	Fail	Fixed by implementing '.reset_index()' and explicit top 5 filtering logic
04	22/04/2026	Shamla Aqthar	Hovering over Sri Lanka correctly displayed the country name and new displacement values in a tooltip.	Pass	
05	22/04/2026	Shamla Aqthar	The most affected country KPI perfectly matched the first bar in the top 10 bar chart.	Pass	

Technical research

Build a Streamlit Dashboard app in Python (2026) Youtube. Available at:

https://youtu.be/p2pXpcXPoGk?si=NBl_UOEIX-aM8y0g (Accessed: 18 April 2026).

st.warning - Streamlit Docs (2025) Streamlit.io. Available at:

<https://docs.streamlit.io/develop/api-reference/status/st.warning> (Accessed: 26 April 2026).